



Summer Internship Task — Offensive Security (Tool Development)

Task Title: Custom Reconnaissance Tool Development

Objective:

Design and develop a lightweight, modular reconnaissance tool to automate initial information gathering during penetration testing engagements. The goal is to familiarize interns with offensive tooling techniques, scripting, and modular architecture in real-world red team scenarios.

Task Description:

Interns are required to develop a CLI-based or lightweight GUI tool that integrates basic recon functionalities. The tool should be written in Python or Go and perform at least the following tasks:

1. Passive Recon:

- WHOIS lookup
- DNS enumeration (A, MX, TXT, NS records)
- Subdomain enumeration using external APIs (e.g., crt.sh, AlienVault OTX, etc.)

2. Active Recon:

- Port scanning (via Nmap wrapper or sockets)
 - Banner grabbing

- Detecting technologies (e.g., using `whatweb`, `wappalyzer` APIs)

3. Reporting:

- Generate a summary report in `.txt` or `.html` format with all gathered information.
- Include timestamps and IP resolution details.

4. Modularity:

- Each module should be independent and callable via command-line flags (e.g., `-dns`, `--whois`, `--subdomains`).
- Logging should be implemented with verbosity levels.

☒ Deliverables:

- Fully functional script/tool with a `README.md` and usage guide.
- Sample report for a test domain (e.g., `example.com`)
- Code must be documented and pushed to a GitHub repository (private or public).
- Bonus: Dockerize the tool for easy deployment.

Deadline:

10th July 2026.

References (Highly Recommended):

- OWASP Recon Guide
- [Python Recon Tools](#)
- [Sublist3r \(for subdomain enumeration\)](#)
- [Go Enumeration Toolkit](#)

Evaluation Criteria:

- **Innovation and Creativity:** Originality of the tool.
- **Technical Execution:** Code quality and functionality.
- **Documentation:** Clarity and comprehensiveness.
- **Team Collaboration:** Effectiveness of teamwork.
- **Final Product:** Usability and impact.

Conclusion:

This task provides a hands-on opportunity to develop a practical security tool. Your work will not only showcase your technical skills but could also contribute significantly to real-world security practices.

Good luck!

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